- *Frozen*—Similar to the cold phase, where the index is switched off for indexing but querying is allowed. However, queries are more infrequent or even rare. When the index is in this phase, users may notice longer query response times.
- *Delete*—This is the index's final stage, where it is deleted permanently. As such, the data is erased and resources are freed. Usually we take a snapshot of the index before deleting it so the data from the snapshot can be restored in the future.

Transitioning from the hot phase to every other phase is optional. That is, once created in the hot phase, an index can remain in that phase or transition to any of the other four phases. In the following sections, we check out a few examples that set an indexing lifecycle policy so the system can manage indexes automatically.

6.9.2 Managing the index lifecycle manually

So far, we've created or deleted an index on demand when needed (intervening manually). But we cannot delete, roll over, or shrink indexes based on conditions such as the size of the index exceeding a certain threshold, after a certain number of days, and so forth. We can use ILM to help us set up this feature.

Elasticsearch exposes an API for working with the index lifecycle policy with the `_ilm` endpoint. The format is `_ilm/policy/<index_policy_name>`. The process is split into two steps: defining a lifecycle policy and associating that policy with an index for execution.

STEP 1: DEFINING A LIFECYCLE POLICY

The first step is to define a lifecycle policy, where we provide the required phases and set the relevant actions on those phases.

Listing 6.48 Creating a policy with hot and delete phases

```
PUT _ilm/policy/hot_delete_policy
{
  "policy": {
    "phases": {
      "hot": {
        "min_age": "1d",
        "actions": {
          "set_priority": {
            "priority": 250
          }
        }
      },
      "delete": {
        "actions": {
          "delete" : { }
        }
      }
    }
  }
}
```

The ILM API

Defines the policy and its phases

Defines the first phase (hot)

Sets the minimum age before other actions are carried out

Sets the priority

Defines the actions that must be carried out

Defines the delete phase